Week 6 Assignment — Fine-Tuning BERT for Real-World NLP Impact

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Abstract

Toxic comment classification is a critical problem in online content moderation, where identifying harmful or abusive language is essential to maintain safe digital environments. This study applies a transformer-based model, BERT, to classify comments as toxic or non-toxic using the Jigsaw dataset. A sample of 15,000 comments was used, and the model was fine-tuned using Hugging Face's Trainer API. The results demonstrate that BERT significantly outperforms traditional models such as Logistic Regression, achieving high accuracy and improved F1-score for the minority toxic class. The findings highlight the importance of contextual understanding in natural language processing and show how advanced models can enhance real-world moderation systems by improving detection accuracy and scalability (Devlin et al., 2019).

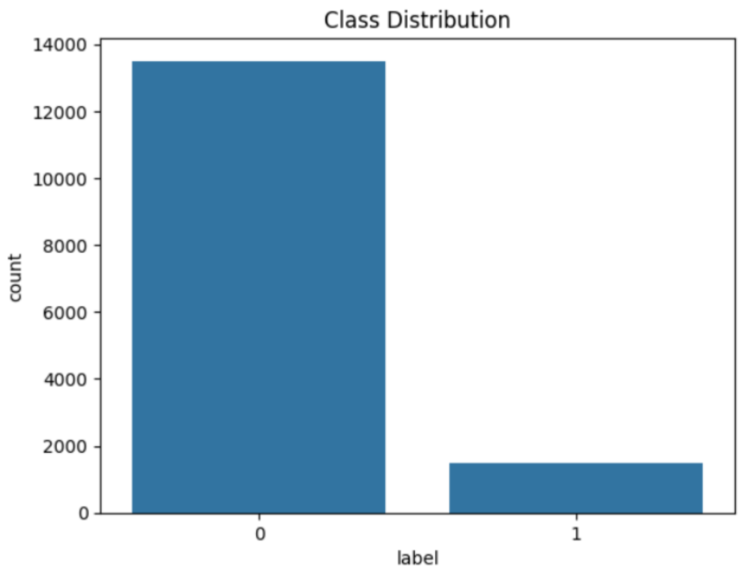
Problem & Objective

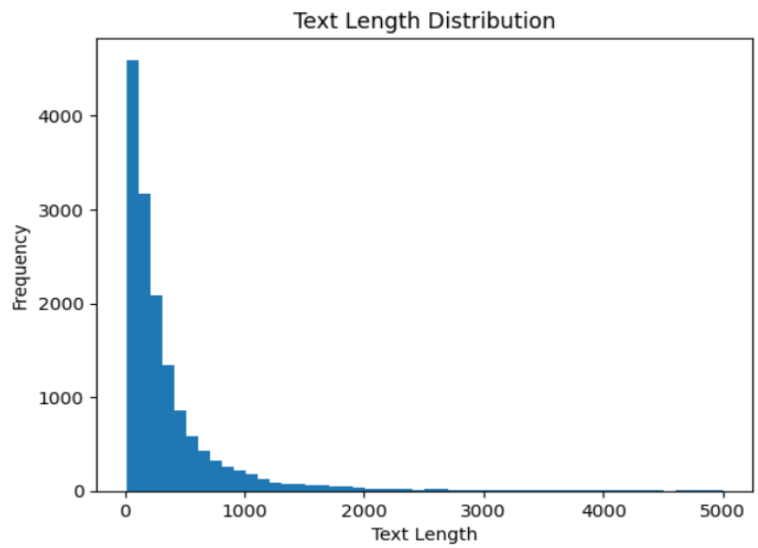
Online platforms face increasing challenges in moderating toxic content due to the complexity of language, including sarcasm, implicit meaning, and contextual dependencies. Traditional machine learning models struggle to capture these nuances (Vaswani et al., 2017). The objective of this study is to fine-tune a pre-trained BERT model for toxic comment classification and evaluate its effectiveness compared to a baseline model. The study also aims to connect technical performance improvements to real-world business impact.

Data & Exploratory Data Analysis (EDA)

The dataset used is the Jigsaw Toxic Comment Classification dataset (Kaggle, n.d.). From the original dataset of over 150,000 comments, a subset of 15,000 comments was selected for efficient training. The dataset exhibits class imbalance, with approximately

90% non-toxic and 10% toxic comments. Exploratory analysis shows that text length is highly skewed, with most comments being short and a few outliers being very long. Additionally, text length alone does not distinguish toxic from non-toxic comments, highlighting the need for models capable of understanding context. A Logistic Regression baseline achieved high accuracy but failed to capture contextual meaning effectively.



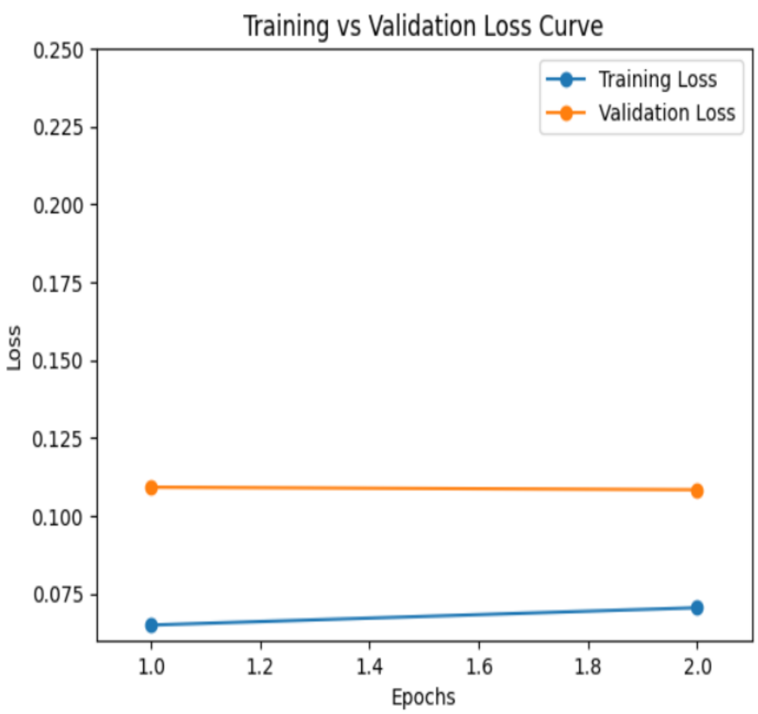


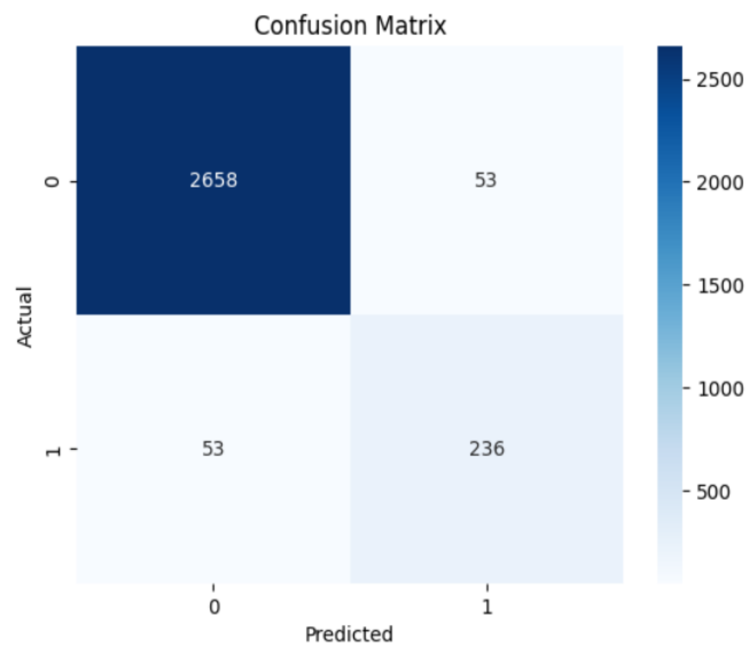
Modeling Approach

The BERT (bert-base-uncased) model (Devlin et al., 2019) was fine-tuned for binary classification using Hugging Face Transformers. Tokenization was performed with padding and truncation, and attention masks were used to preserve contextual relationships. The model was trained for two epochs using the AdamW optimizer with a learning rate of 2e-5. Training and validation datasets were split using an 80/20 ratio. The learning rate schedule decreased over training steps, improving stability and convergence. BERT uses self-attention mechanisms to analyze relationships between all words in a sentence simultaneously, enabling better contextual understanding compared to traditional models.

Results & Comparison

The fine-tuned BERT model achieved an accuracy of approximately 96.5% and an F1-score of 0.817, demonstrating strong performance on the minority toxic class. In comparison, the Logistic Regression baseline achieved around 93.9% accuracy but showed weaker performance in identifying toxic comments due to its inability to capture contextual meaning. Training loss decreased steadily, while validation loss stabilized around 0.108, indicating proper convergence and minimal overfitting. The confusion matrix further confirms that most predictions are correct, with relatively low false positives and false negatives.





Business Impact

The improved performance of BERT directly translates into real-world value for content moderation systems. A recall of 0.817 means that approximately 82% of toxic comments are correctly identified, reducing exposure to harmful content. High precision ensures that non-toxic comments are not incorrectly flagged, minimizing user dissatisfaction. These improvements reduce manual moderation workload, enable real-time filtering at scale, and enhance user trust on digital platforms. Organizations can leverage such models to improve operational efficiency and maintain safer online communities. In large-scale platforms handling millions of comments daily, even a 5–10% improvement in toxic content detection can significantly reduce manual moderation effort and operational costs. This enables faster response times and improves overall platform safety and user trust.

Limitations

Despite strong performance, several limitations exist. The dataset is imbalanced, which may bias predictions toward the majority class. The model operates on individual comments and does not incorporate conversational context. Additionally, potential bias in training data may lead to unfair predictions for certain demographic groups. Future improvements can include techniques such as class weighting or resampling to better handle class imbalance and improve minority class detection.

Conclusion

This study demonstrates that fine-tuning BERT significantly improves toxic comment classification compared to traditional approaches. The model effectively captures contextual relationships, leading to better detection of harmful language. The results highlight the importance of transformer-based architectures in modern NLP tasks and their ability to deliver meaningful business impact. Future work can further enhance performance by addressing dataset limitations and expanding model capabilities.

References

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